



# Probabilistic Bayesian perceptron basis function with convolutional Gabor component regression in scalability analysis for structure health monitoring system: a machine learning algorithms

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## Abstract

Conventional procedures in field of structural health monitoring (SHM) are typically time-consuming, complex, and harmful. To properly assess and track the structural integrity, a wide range of sensors is also required. Thus, machine learning techniques are giving SHM systems the tools they need to improve their skills and offer clever answers to historical problems. This research proposes novel technique in scalability enhancement in SHM for smart healthcare development using machine learning model. Here the data processing with damage detection is carried put using probabilistic Bayesian perceptron basis function with convolutional Gabor component regression (PBPBF-CBCR) model. Then the processed data segmentation is carried out using stacked quadratic finite element decision tree discriminant analysis. The experimental analysis is carried out in terms of Detection accuracy, Correlation coefficient, reliability, Scalability, Processing time. Proposed technique attained Correlation coefficient of 97%, Detection accuracy of 95%, RELIABILITY of 94%, processing time of 58%, SCALABILITY of 95%.

**Keywords** Scalability enhancement · SHM · Smart healthcare development · Machine learning model · Data processing · Data segmentation

## 1 Introduction

Over the course of their lives, infrastructure is continuously subject to a variety of stimuli, including changes in the environment, traffic, chemicals, and human activity, all of which have a detrimental effect on their conditions. In order to evaluate the operational behaviour of civil structures, identify structural degradation early, avoid catastrophic failures, and prolong their lifespan, it is necessary to conduct routine monitoring. However, because infrastructures are rather huge, the inspection process has been regarded as time-consuming and costly. Ultrasonic, thermography, X-ray, and visual examination are examples of traditional non-destructive evaluation techniques that can yield comprehensive information about the condition of structures. Nevertheless, they necessitate skilled manpower and accessibility, which frequently result in disruptions to operating services (Kosova et al. 2025). Therefore, to enable proactive and cost-efficient maintenance, it is essential to develop effective SHM approaches. For engineering applications, data-driven structural health monitoring (SHM) techniques have recently emerged as a state-of-the-art method that aids in preserving structural integrity and expedites maintenance procedures (Manoni et al. 2024) & (Islam et al. 2024). Because delamination in composite materials occurs internally, standard visual examinations are insufficient, hence sophisticated SHM procedures are required to find and fix these hidden structural weaknesses. Consequently, a variety of non-destructive evaluation methods including SHM approaches have been studied in order to identify deterioration in composite structures (Cha et al. 2024). Despite their efficacy, majority of these strategies need a high level of operator experience to be implemented successfully. These methods are often expensive, time-consuming, and complex. Additional operational disturbances result from the inspection process, which usually calls for the

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composite structures to be out of commission and makes use of specialised signal production equipment. The intrinsic durability benefits of composite buildings are severely compromised by these limitations, making them inappropriate for uses where catastrophic failure is intolerable. As a result, vibration-based monitoring techniques have become more well-liked as a viable and approachable substitute for composites' SHM. By simply adding vibration sensors to the building and examining the subsequent dynamic responses, this technology offers a straightforward, effective, and dependable solution (Azad et al. 2024). However, this sole reliance on system physics is rather impractical in the case of complex real-world structures due to a number of factors, including non-linearity in materials and/or geometry, uncertainty in material properties, boundary conditions, environmental/operational variations, and the difficulty of modelling complex real-world structures. This restricts how these techniques can be used for monitoring the health of relatively simple buildings with well-controlled surroundings and pre-established boundary conditions. The more complex the underlying system, the less reliable such an approach becomes. This encourages the adoption of a data-driven method for SHM, which avoids some of the main issues with the physics-based approach by treating damage assessment, at least at a lower level, primarily as a kind of statistical pattern identification issue (Ong et al. 2024).

The method can be time-consuming and environmentally unfriendly when it comes to composite material, structural monitoring using lab equipment, workshop methods. Nonetheless, machine learning (ML) can carry out this process with a respectable level of accuracy. In general, machine learning (ML) is quicker and less expensive than conventional laboratory material testing when it comes to predicting material qualities. Furthermore, a variety of accurate data on material performance in various scenarios that would be difficult to detect through laboratory trials can be obtained by statistical modelling and simulation. As a result, ML has drawn more attention in a variety of SHM applications, ranging from experimental design instruction to the prediction of fundamental features (Guo et al. 2024). In this area, high-throughput material damage identification and quantification will be possible through the use of algorithmic techniques. ML methods are utilized in field of SHM in previous studies. Nonetheless, it is critical to compare the application of ANNs, CNNs, SVM methods while accounting for input data, feature selection strategies, structure of interest, data size, extent of damage identification, machine learning model's accuracy. Aside from fact that all of the aforementioned ML methods are applied to comparable structural damage as well as system identification tasks, some of them work better when data originates from multi-sensor data fusion or when damage-sensitive feature

extraction techniques, like wavelet packet transform (WPT) or Hilbert-Huang transform (HHT), are applied to the data.

Contribution of this research: to design and construct a scalable and efficient Structural Health Monitoring (SHM) data processing model that leverages multi-modal imaging techniques to reduce processing time. Multi-modal imaging integrates data from various sources such as thermal, infrared, acoustic sensors to enhance accuracy as well as reliability of SHM. By optimizing data handling as well as computational efficiency, proposed model ensures smart healthcare data analysis, enabling quick decision-making and Processing to structural issues in smart cities.

## 2 Related works

To classify the retrieved features, which change from an intact condition to a damaged state, extensive research into conventional machine learning and classification techniques has also been carried out. Artificial neural networks (ANN), Bayesian probabilistic techniques, fuzzy logics, simple genetic algorithms (GA), multi-objective GA, and support vector machines (SVM) are a few of the fundamental and innovative applications. Using a specified sliding window size to localise the identified cracks in RGB images, CNN architecture in was created to identify concrete cracks. The results showed a 97% detection accuracy, which was highly successful. A method for identifying damage using output-only response data was provided in Work (Srinivasan et al. 2024). Measured acceleration response was used to create training data for several simulated damage (loose bolt) scenarios. PoD (Probability of damage) indicator was established following training of distinct CNNs for every damage scenario. It was demonstrated that individual cases were correctly identified with an average inaccuracy of 0.54% when examining cases with undamaged, single damage, multiple damage. In a related work, the acceleration response was recorded using three-dimensional wireless sensors (Han et al. 2024). But this method necessitated a lot of response data linked to different combinations of loose connections, making the application unfeasible. To mitigate the disadvantages, the author (Clero et al. 2025) proposed a novel method that only needed 2 states of damage—undamaged as well as fully-damaged situations. Although computational process was comparable to earlier study, this approach was limited to identifying the structure's general state. A CNN mimicking the FE method of a simply supported beam was trained in Work (Rahman et al. 2024), considering both noisy as well as noise-free states. An undamaged laboratory-scale 3D steel bridge's acceleration signal was used by the author (Kumar and Kota 2024) to suggest an unsupervised technique. A two-dimensional (2D)-CNN autoencoder was used to extract key features from the

response signal vectors after they had been normalised and subjected to CWT as well as FFT. As novelty-detectors that matched the sensors, One-Class Support Vector Machines (OC-SVM) were employed. Studies by Renugadevi et al. (2024) have shown that WPT (wavelet packet transform) of vibration signals and visual data are effective in localising damage. Robust damage localisation as well as damage intensity estimation can be achieved with a CNN configuration that operates in parallel. One-dimensional (1D) CNN is used to process time-domain data, while 2D CNN (lower branch) is used to process WPT or vision data. Feature maps are then concatenated for classification or regression. The author (Forlesi et al. 2024) provided a thorough analysis of the body of research on IoT-based SHM frameworks. The significance of SHM and ways in which IoT technology can improve SHM systems are covered by the writers. They go over the several IoT-based sensors as well as gadgets that are utilised for SHM as well as various methods for gathering and analysing data. Additionally, the authors list the drawbacks and restrictions of IoT-based SHM systems as well as possible ways to overcome them. Overall, by outlining the state-of-the-art as well as potential future research avenues in this area, this study offers researchers as well as practitioners working on IoT-based SHM systems a useful resource. A thorough analysis of the most recent advancements in DL based SHM of civil infrastructures was given in work (Chen et al. 2024). The study examines the many kinds of DL methods that are created for civil infrastructure monitoring and talks about advantages of applying DL methods, such as ANN as well as CNN, for SHM applications. An overview of application of IoT for SHM of civil engineering structures was provided by author (Armijo and Zamora-Sánchez 2024). Writers go over the advantages and difficulties of utilising IoT technologies for SHM, such as enhanced connectivity, data collecting, and analysis. In order to illustrate potential advantages of these methods in enhancing safety as well as dependability of civil engineering structures, article also offers case studies of IoT-based SHM in various civil engineering structures. Similarly, in order to visually examine high voltage transmission lines with more accuracy, (Plevris and Papazafeiropoulos 2024) created a remote distance inspection method based on laser ranging as well as image processing technologies. A high-resolution camera, a computer, 2 laser rangefinders, a digital compass, GPS are all used in the created system to take pictures and gather data on object's location, height, distance from transmission line (Mardanshahi et al. 2025).

To classify the insulator outer surface that represents hydrophobicity, they retrieved co-occurrence matrix-based features as well as used them as input to create a polynomial classifier. For given problem, suggested method effectively obtained a 97% categorisation rate (Yang and Huang 2024). Because the suggested method relies solely on intensity and

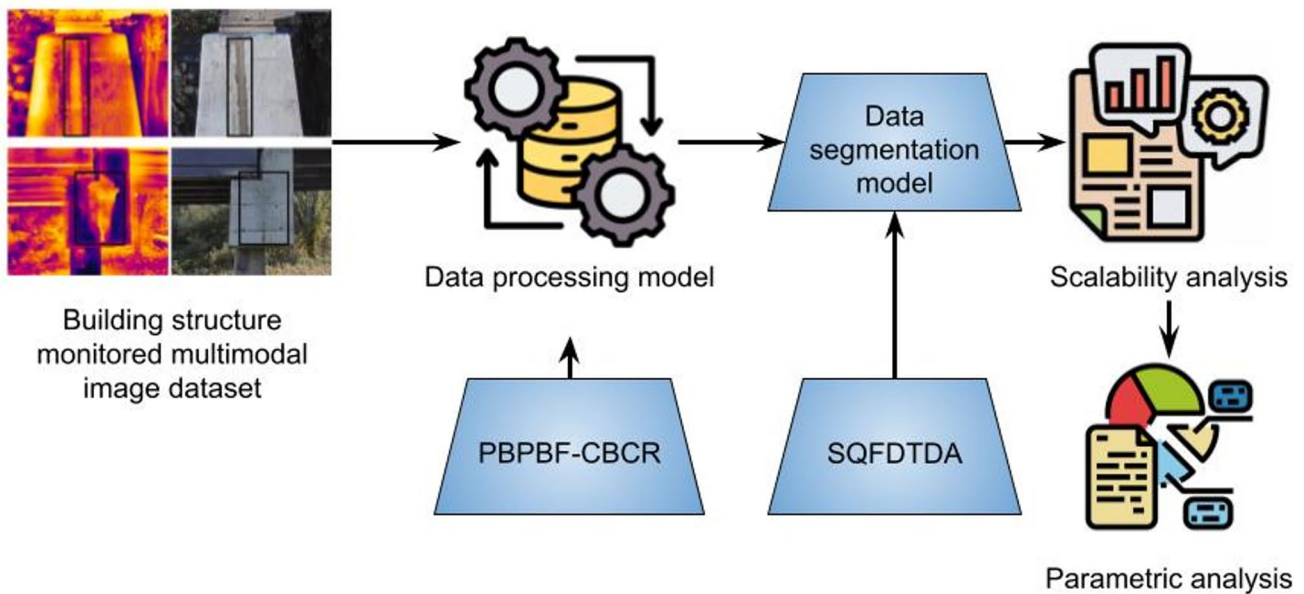
statistical characteristics, it was unable to be used to detect anomalous temperature distribution across the insulator surface. However, the authors were able to identify snow as well as water droplets on insulator surface with an 87% recognition accuracy. A thermal image is often produced as a result of the anomalous temperature distribution caused by the current discharging event via the insulator surface (Mudabir et al. 2025). For example, (Yu et al. 2024) created an image tracking technology-based system for visual inspection of insulators and fault monitoring utilizing infrared images. In order to construct a fault diagnosis system, they used the Otsu threshold segmentation approach and histogram in their study to determine the relative temperature of the insulator. However, the usefulness of the suggested approach for wider use in real-world applications is limited because it was only tested on 25 infrared photos. In order to show the effectiveness of infrared and ultraviolet (UV) imaging methods for detecting faults in composite insulators caused by corona discharge as well as heating linked to leakage current flow, a series of tests was conducted in the University of Natal's high voltage laboratory in Okur et al. (2025). The findings suggested that both methods be applied when manually inspecting composite insulators in electricity sub-stations. Used the watershed city block transform technique to segment the image in order to identify faults in IRT images (Rajeswari et al. 2024).

### 3 Proposed scalable and efficient structural health monitoring (SHM) model

An effective non-invasive visual examination system is shown in this section. The following are the primary steps that make up the suggested system: Image processing (i), region of interest extraction and segmentation (ii), and PBPBF-CBCR\_SQFDTDA creation (iii). Proposed method block diagram is given in Fig. 1. Class labels may or may not be applied to data at this point, depending on the information gathered. Then, these data undergo pre-processing to minimise the size of the damage vectors and eliminate any noise or outliers. Feature extraction is the next step. Damage-sensitive features are chosen at this stage using either engineering judgement, mathematics and transformation techniques, or a mix of two. Following feature extraction, postprocessing can be used to further compress, normalise, or fuse data as necessary.

#### 3.1 Data pre-processing

A significant amount of data is generated by thermal infrared cameras. For reading reasons, the DHU Night Dataset is first split into 10 sections based on participants' types



**Fig. 1** Proposed architecture of PBPBF-CBCR\_SQFDTDA

and various activities. The samples are then processed by tacking out data at a rate of 100 Hz. Distinct values were then depended upon by modes and users. Consequently, any faulty data that were deemed false at input stage were subjected to NN. Since it doesn't appear to have much of an impact on outcome, we have simply removed User's name column from the majority of models in order to address this. For better outcomes, data characteristics are stored independently and integrated in the code when necessary. The MHAD Dataset likewise uses all of the aforementioned procedures.

The measurement, data collection, and data preparation were done using an infrared thermal imaging instrument. The image's temperature span can be changed to make it easier to identify the subject in situations where the background is busy and there is a lot of outside influence. Following import, the MATLAB built-in function `grayscale` and `binarizes` the image, using Otsu's approach to determine threshold for minimising intraclass variation of black and white pixels. In order to convert an intensity image to a binary image, function calculates a global threshold. The image's noise may be eliminated to achieve the structural processing because of the stark temperature difference between the backdrop and the heat sources. In particular, the imager's temperature range was set between 26 and 35 °C throughout the current study's tests, whereas room temperature is approximately 25 °C as well as temperature of the heat resource is 30 °C. Every pixel in a colour image has a greyscale value after it has been converted to greyscale. Due to significant temperature contrast in vicinity of heat sources, a global threshold set utilizing previously indicated

function is useful for binarization; as a result, the noise in the region of the heat sources is removed. Then, to eliminate noise far from heat sources, we crop the photos so that only the heat source area is left. Furthermore, during binarization, voids usually form inside the heat source region due to the unequal heat sources; consequently, 8-connectivity is used to fix the flaws. Two adjacent pixels are regarded as a region if they are connected in transverse, longitudinal, diagonal directions. Full area of heat source is marked in 2D image by filling each pixel inside region with value 0. Following acquisition of target's characteristics, centre of heat source region is used to determine target's displacement data in each snapshot.

### 3.2 Probabilistic Bayesian perceptron basis function with convolutional gabor component regression (PBPBF-CBCR)

A statistical multivariate method for a set of variables  $X = X_1 \dots X_n$  is called a Bayesian network (BN). They can be characterised by two elements and were created for the depiction of knowledge in the face of uncertainty: Qualitative: the (in) dependence encies between variables are determined by arcs connecting nodes in a directed acyclic graph. Quantitative: given its parents in graph, represented by  $pa(X_i)$ , which are  $i \in 1, 2, \dots, n$ , each variable  $X_i$  has a conditional distribution  $p(X_i | pa(X_i))$ . BN was suggested for variables that are discrete. The literature contains a comprehensive and well-organised theory and methodology. Nonetheless, continuous domains are frequently seen in environmental data. Discreteizing the continuous variables



and treating them as though they were discrete is the most popular approach in this situation. BN also permits the inclusion of newly learnt data, or evidence, in the model through probabilistic propagation, often known as the inference process. The inference procedure entails computing posterior distribution  $p(X_i|e)$  for every variable of interest in  $X_i \in E$ , assuming that set of evidence variables is represented as  $E$  as well as that their values are represented as  $e$  is given by Eq. (1)

$$p(x_i | e) = \frac{p(x_i, e)}{p(e)} \propto p(x_i, e) \quad (1)$$

The constant  $p(e)$  holds for any  $X_i \in E$ . Thus, the following method of computing as well as normalising marginal probabilities  $p(x_i, e)$  can be used to carry out this process is given by Eq. (2)

$$p(x_i, e) = \sum_{x \in \{x_i, e\}} p_e(x_1, \dots, x_n) \quad (2)$$

where the probability function  $p_e(x_1, \dots, x_n)$  is the result of substituting the values of the evidence variables  $E$  in  $p(x_1, \dots, x_n)$  by their values. If a BN has a continuous response variable ( $Y$ ) as well as collection of discrete and/or continuous feature variables ( $X_1; \dots; X_n$ ), it can be utilised as a regression method for prediction. Therefore, the conditional density is calculated to forecast value for  $Y$  from  $k$  observed features, numerical prediction for  $Y$  is provided utilizing expected value in the manner described below (3)

$$y = g(x_1, \dots, x_n) = E[Y|x_1, \dots, x_n] = \int_{\Omega_Y} y f(y|x_1, \dots, x_n) dy \quad (3)$$

where  $Y$ 's domain is denoted by  $\Omega_Y$ . Consider conditional density  $f(x_i|y)$  for two random variables,  $X_i$  and  $Y$ . The plan is to use equal frequency approach with three intervals to divide domain of  $Y$ . The exponential function's concave and domain-wide increasing features are then used to separate the domain of  $X_i$ . As a result, the partition is made up of a sequence of intervals whose boundaries match locations where empirical density shifts from concavity to convexity, or from decrease to rise. Every split of support of  $X$  has a five-parameter MTE fitted to it, meaning that five parameters must be calculated from the data in each split is given by Eq. (4)

$$f(x) = a_0 + a_1 e^{\alpha_2 x} + a_3 e^{a_4 x}, a < x < \beta, \quad (4)$$

where the interval used to estimate the density is defined by  $\alpha$  and  $\beta$ . Calculating the distribution's expectation is a

logical method of obtaining the expected value. Therefore, using a density described as in Eq. (5), the anticipated value of a random variable  $X$  is calculated as

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x f(x) dx \\ &= \int_{\alpha}^{\beta} x (a_0 + a_1 e^{a_2 x} + a_3 e^{a_4 x}) dx \\ &= a_0 \frac{\beta^2 - \alpha^2}{2} \\ &\quad + \frac{a_1}{a_2^2} ((a_2 \beta - 1) e^{a_2 \beta} - (a_2 \alpha - 1) e^{a_2 \alpha}) \\ &\quad + \frac{a_3}{a_4^2} ((a_4 \beta - 1) e^{a_4 \beta} - (a_4 \alpha - 1) e^{a_4 \alpha}) \end{aligned} \quad (5)$$

Expected value is sum of aforementioned expressions for every portion if density was defined by distinct intervals. A fundamental issue in SHM is reasoning as well as decision-making under uncertainty, PGMs offer a strong foundation for doing so. Joint probability distributions are represented using probabilistic graphical models, where nodes indicate a collection of random variables and edges between nodes suggest that variables are dependent on one another. The graph's representation of joint probability distribution is derived by Eq. (6):

$$P(X_1, X_2, \dots, X_N) = K \prod_{i=1}^M f(Y_i) \quad (6)$$

where  $K$  serves as a normalisation factor to guarantee that sum of the probabilities equals one. Up until  $L$ th level of hierarchy, which is made up entirely of base units, one can keep expanding the substructures into constituent units as one moves down the hierarchy levels. Each unit within a structure is uniquely identified by notation  $uni_i$ , where  $i$  is an integer from 1 to  $L$  as well as  $n$  is an integer from 1 to  $N_{ui}$ .

Structure  $S$  also has a predicted time-dependent health state vector  $H(t) = \{hc_1^2(t), \dots, hc_{N_c}^L(t), hj_1^2(t), \dots, hj_{N_j}^k(t)\}$  any subset of joints and components can be used to create health-state vectors.

A set of failure modes of interest  $F = F_1, \dots, F_N$  must exist for the structure/system  $S$  in order for it to stop being suitable for its intended use. A certain failure mode is thought to be reliant on health conditions of a subset of joints, substructures, components for which a health-state vector is created. Additionally, there is a utility  $U_{Fn}$  linked with each failure state. Lastly, a set of decisions  $d = d_1, \dots, d_{Nd}$  that impact  $H^*$  for the structure  $S$  also exist; each decision has a corresponding utility  $U_{di}$ . Furthermore, the

distribution of  $v$  will change due to a set of environmental parameters  $e = e_1(t), \dots, e_n(t)$ .

An input layer, a hidden layer, output layer are 3 layers that typically make up an MLP. The feed-forward ANN class includes the early MLP. Similar to the perceptron, the early MLP uses error corrective learning to calculate the connection weight between two layers. This learning modifies the connection weight based on discrepancy between expected as well as actual output. Error corrective learning, however, is not cross-layer compatible. Expected output for a training sample in a supervised learning scheme is denoted as  $y(l) \in \mathbb{R}^{P \times 1}$ , where  $P$  is number of outputs. Mapping function  $F: x(l) \rightarrow y(l)$  is NN goal.

This mapping is accomplished during the training process, where a set of sample pairs  $(X(l), Y(l))$  are used to update the network's parameters  $\Theta_{NN}$  through a minimisation issue of a loss function  $L(\Theta_{NN}) = L(F(X, \Theta_{NN}) - Y)$ . Regression issues employ half-mean-squared error as a loss function. The activation function, which might take the shape of sigmoid, ReLU, leaky ReLU, or tanh functions, is in charge of introducing non-linearity into the system is given by Eq. (7)

$$x_{i+1,k} = T \left( \sum_{j=1}^m w_{ikj} x_{ij} + b_{ik} \right) \quad (7)$$

The network is fed labelled input data during the training phase, and its internal parameters are changed to reduce the discrepancy between true labels and anticipated outputs. Due to the multivariate optimisation involved in this process,  $\Theta_{NN}$  is not always unique and can generate many networks that fit the data.

A regularisation loss is often added in the form of  $R(\Theta_{NN}) = \lambda \sum_{i,j} \Theta_{ij}^2$ , where  $\lambda$  is a regularisation parameter, to convey the preference for specific weight sets and eliminate some of this ambiguity. L2 regularisation is term utilized to describe this process, which causes optimisation algorithm to choose models with fewer parameters. Because highly specialised networks with concentrated parameters are penalised more than those with a more uniform weight distribution, this regularisation technique also has an effect on lowering overfitting.

#### 4 Convolutional Gabor component regression

Convolution is the initial stage, in which the input feature map is convolved with a filter or kernel. In mathematics, this operation is expressed as  $Y = X * W$ , where  $Y$  is output feature map that results from convolving input  $X$  with kernel  $W$ . An activated feature map  $Z$  is produced by applying

an activation function element-wise to introduce non-linearity following the convolution procedure. Pooling procedure, which comes next in a CNN, helps to minimise feature map's spatial dimensions while maintaining crucial data.

Typically, a series of convolution+pooling layers are used to extract increasingly complicated characteristics from input. To solve issue of internal covariate shift, normalisation layers, like Batch Normalisation, can be employed throughout the network to normalise the activations inside a layer.

Normalisation layers facilitate faster convergence, enhance the network's capacity for generalisation, and stabilise and regularise the learning process by normalising the activations. Additionally, some neurones can be randomly dropped to zero during training by using a dropout layer.

Every Gabor filter has a  $S$  scale and a  $D$  direction. Scale of feature maps is embedded into various layers; feature maps are encoded in learnt layers in order to incorporate the Gabor filter into the CNN. By using Gabor filters to capture direction and scale, CNN's learning capacity to comprehend the information is enhanced. The collection of weights in a conventional CNN is formed as a tensor  $(y_{out}, y_{in}, W, W)$ , where  $y_{in}$ ,  $y_{out}$ ,  $k_{wi}$ , and  $k_{li}$  stand for input feature map, output feature map, convolutional kernel width, length. The weights in GCNN are redesigned as  $(y_{out}, y_{in}, D, k_{wi}, k_{le})$ . To create an intrinsic set of convolutional layers based on the Gabor filter, quantity of feature map must be maintained throughout the convolution process is given by Eq. (8)

$$f_i(d, s) = f_i^{out} \odot G(d, s) \quad (8)$$

where a collection of Gabor filters with distinct  $d$ ,  $d = (1, \dots, D)$  directions and  $s$ ,  $s = (1, \dots, S)$  scales, learnt kernel are represented by  $f_i(d, s)$ ,  $f_{out\ i}$ , and  $G(d, s)$ . Element-wise product operation between real part of Gabor filters is represented by symbol  $\odot$ . The definition of Gabor filter set is given by Eq. (9)

$$G(d, s) = \frac{\|k(d, s)\|}{4\pi^2} \exp \left( -\frac{\|k(d, s)\|^2 \|z\|^2}{8\pi^2} \right) \left( \exp(jk(d, s)z) - \exp(2\pi^2) \right) \quad (9)$$

where  $z = (X, Y)$  and  $k(d, s)$  is described as Eq. (10)

$$k(d, s) = \frac{\left(\frac{\pi}{2}\right)}{\sqrt{2}^{(p-1)}} \exp \left( jd \frac{\pi}{D} \right) \quad (10)$$

Consequently,  $f_i(d) = (f_i(d, 1), \dots, f_i(d, S))$  is the modulated Gabor convolutional network. Although feature maps with dimensions of  $y_{out}$ ,  $y_{in}$ ,  $D$ ,  $k_{wi}$ , and  $k_{le}$  are necessary for every Gabor filter-based convolutional layer, only

feature maps with these dimensions are stored as a result of the Gabor filter's use.

As a result, Gabor filter is built into CNN by lowering network specification, which lessens the computational load on suggested approach while also improving learning. To accomplish PCA, we can apply these constraints to both weight matrix  $W_a$  and  $W_c = WT$  during training stage. Two simultaneous operations are added by the encoding module.

The linear transformation is represented by  $F_a(m; \theta_a)$ . The feedforward architecture 24 of the residual connection  $F_b(m; \theta_b)$  applies a linear transformation through  $L_b$  layers, followed by a nonlinear activation function  $g$ . Lower-dimensional vector  $z$  is thus obtained by Eq. (11)

$$z = F_E(m) = F_a(m; \theta_a) + F_b(m; \theta_b) \quad (11)$$

Decoding step  $F_D$ , which recreates original measurement from its lower dimensional representation, follows the same architecture is given by Eq. (12)

$$\hat{m} = F_D(z) = F_c(z; \theta_c) + F_d(z; \theta_d) \quad (12)$$

The linear transformation is applied here via the operator  $F_c(z; \theta_c)$ , and the residual link with feed forward network architecture as well as nonlinear activation functions is  $F_d(z; \theta_d)$ .

#### 4.1 Stacked quadratic finite element decision tree discriminant analysis (SQFDTDA)

Encoding and decoding are the two steps that make up the fundamental processes of autoencoders. Each observation  $x \in X^{n \times m}$  is converted into a coded representation  $h$  by an encoding operation  $\phi: R^m \rightarrow R^d$ , where  $n$  is number of observations,  $m$  is number of features,  $d$  is number of retained components. This encoding process is described as a forward propagation process.

Through a decoding process  $\psi: R^d \rightarrow R^m$ , the input data from  $h$  is reconstructed as a vector  $\hat{x}$ , with a corresponding residual error  $e = \|x - \hat{x}\|$ . Projected residual error is nearly negligible if network is successful in completing this encoding/decoding procedure. Finding set of specifications  $\Theta = \{\theta, \theta'\}$  that minimises loss function is the formal goal of the simplest learning process is given by Eq. (13)

$$Loss(\Theta) = \frac{1}{n} \sum_{i=1}^n \|x_i - g_{\theta'}(h_{\theta}(x_i))\|^2 \quad (13)$$

where  $x_i$  is  $i$ th observation from matrix  $X$ , and  $h_{\theta}(\cdot)$  and  $g_{\theta'}(\cdot)$  are the activation functions for encoding as well as decoding layers. The degree of dissimilarity between  $g_{\theta'}(h_{\theta}(x_i))$ , or

reconstructed input data  $\hat{x}_i$ ,  $x_i$  is measured by this averaged reconstruction error function. Sigmoid functions are typically incorporated in encoding and decoding layers is given by Eq. (14)

$$\begin{aligned} h_{\theta}^{(k)}(x_i) &= f\left(\sum_{j=1}^m \theta_{k,j} x_i^{(j)} + b_k\right) \\ g_{\theta'}^{(k)}(h_i) &= f\left(\sum_{j=1}^d \theta'_{k,j} h_i^{(j)} + b_k\right) \\ f(q) &= \frac{1}{1 + e^{xp-q}} \end{aligned} \quad (14)$$

Consequently, set of model parameters becomes  $\Theta = \{\theta, b\}$ . The term “symmetric architecture” is frequently used to describe this design. Multiple autoencoders are trained consecutively to create an SAE, where output of one layer is sent as input to another autoencoder, to learn more robust and abstract features.

For encoding and decoding tasks, resulting network may consist of several layers, forming a deep architecture. According to the features seen in the training data, this type of network is sparse, which produces specialised neurones. As a result, only when the right kind of stimulus is applied to the input does a neurone provide some information on its output. Every neurone is limited to reacting to a certain feature that appears in a tiny portion of the training instances. Equation (15) can be made sparser by including a constraint term. The most widely used sparsity regularisation term for SAEs is the Kullback–Leibler divergence, which is provided by

$$\begin{aligned} \sum_{k=1}^l D_{KL}(\rho || \hat{\rho}_k) &= \sum_{k=1}^l \rho \log\left(\frac{\rho}{\hat{\rho}_k}\right) \\ &\quad + (1 - \rho) \log\left(\frac{1 - \rho}{1 - \hat{\rho}_k}\right) \\ \hat{\rho}_k &= \frac{1}{n} \sum_{i=1}^n [f_k(x_i)] \end{aligned} \quad (15)$$

where  $f_k$  is activation function for  $k$ th neurone,  $l$  is number of hidden units,  $\rho$  is its anticipated value,  $\hat{\rho}_k$  is average activation value of  $k$ th neurone across whole training set. The similarity between two distributions is measured by this cost function, which is higher as they diverge from one another and becomes zero when  $\hat{\rho}_k$  goes to  $\rho$ . The following are the FE equations of motion that need to be resolved by Eq. (16)

$$M(\rho)A + C(K, M, \alpha, \beta)V + K(E)U = F \quad (16)$$

where global acceleration, velocity, displacement vectors are denoted by  $A$ ,  $V$ ,  $U$ . Global mass, damping, stiffness matrices of structure are given by  $M$ ,  $C$ , and  $K$ . These matrices are dependent on method physical specifications of elasticity  $E$ , density  $\rho$ , and damping  $\alpha$  and  $\beta$ . It is thus possible to replicate the acceleration time series that will be used for SHM learning for arbitrary load instances, sampling the parameters from a selected random distribution. Each parameter's upper and lower bounds for random sampling are user-estimated using the FE models' expected model errors. Since uncertainty bounds estimation is not the main topic of this study, users may consult more relevant works that combine uncertainty limits estimation and FE data production. A high sensitivity to localised damage can be enabled by utilising the derivatives of the mode shape. Central difference estimate to 2nd derivative of mode shape vector  $\phi$  is used to get mode shape curvature  $\kappa$  is given by Eq. (17)

$$\kappa_i = \frac{\varphi_{i-1} - 2\varphi_i + \varphi_{i+1}}{h^2} \quad (17)$$

where  $h$  is distance between sensor locations  $i$ , which are equally spaced. Both forward and backward difference approximations are used to estimate the curvature at the ends. Multiple modes can be considered thanks to summed absolute mode shape curvature difference, which is defined as Eq. (18)

$$\Delta\kappa = \sum_{j=1}^R |\kappa_{ju} - \kappa_{jd}| \quad (18)$$

where  $R$  represents total number of modes considered. Additionally, the damaged and undamaged structures are denoted by the subscripts  $d$  and  $u$ , respectively. To align the experimental and numerical data, the cumulative absolute mode shape curvature difference is normalised between 0 and 1 before being fed into the machine learning model. Any damage-induced discontinuities in the mode forms are

amplified by the derivative process associated with the curvature of the mode shapes.

Initially, local sensor setup (SG1) is subjected to mode shape curvature, which is derived from a beam. To determine whole formulas for every sensor, setup is separated into four sections along the bridge's longitudinal axis. Each part's formulation is applied independently before being concatenated. Second, for both experimental (test) and numerical (training) data, the bridge's undamaged state is represented by FE model.

Considering SHM method is implemented at any point throughout bridge's lifetime is the same as this illustration. This method works well for bridges that are already in place, but it needs an accurate FE model. 20% of training data is utilized for testing, remaining 80% is utilized for training.

Training data is subjected to a grid search with fivefold cross-validation to determine ideal hyperparameters and improve suggested algorithm's capacity to generalise to new data. As most crucial hyperparameters of suggested technique, regularisation parameter  $C$ , kernel functions, associated kernel coefficients  $\gamma$  are utilised as input in the grid search. To find the optimal cross-validation score, the grid search does a thorough search across the designated hyperparameters. Additionally, the hyperparameter combination that yields the highest cross-validation score is selected appropriately.

## 5 Results and discussion

**Simulation setup**-High end computer with two Geforce GTX 1080Ti GPUs, 64-GB of RAM, and an Intel Xeon CPU with 16 cores was used for all of computations. Experimental setup is shown in Table 1.

**Training and Validation**- 25% of remaining data are also utilised as test data, and 75% of the data were used for training using Python on a GPU. We changed the batch size, epochs, activation function, loss function, number of hidden layers, neurones in every hidden layer, epochs during training. The data's  $x$ ,  $y$ ,  $z$  coordinates are normalised, sigmoid, ReLu, softmax are used as activation functions.

**Dataset description:** Data are a fundamental component of all SHM data-driven methodologies. Along with development of DL-based methods that can analyse as well as interpret vast amounts of data, significance of trustworthy data for SHM applications has been clear in recent years. Although there isn't yet a standard, open-access dataset for DL-based SHM, people have produced datasets for a few applications in lab-controlled settings. For instance, a database named SDNET2018 was developed that includes 56,000 images of a bridge deck, wall, pavement as well as is appropriate for machine learning applications. Images from

**Table 1** Experimental setup

| Item                  | Contents                                          |
|-----------------------|---------------------------------------------------|
| Processor             | AMD Ryzen 7 4800H with Radeon Graphics @ 2.90 GHz |
| Solid state hard disk | 512 GB                                            |
| Operating system      | Windows 10                                        |
| Memory                | 16 GB                                             |
| Keras                 | Version 2.5                                       |
| GPU                   | NVIDIA GeForce RTX 2060                           |
| Software              | PyCharm community edition 2020.3.2                |
| Study rate            | 0.01                                              |
| Momentum              | 0.9                                               |



a variety of structures, including buildings, bridges, dams, pavements, laboratory testing, are included in dataset. Augmentation techniques, which address camera position and image quality as well as light and weather conditions, can be used for tiny datasets. These techniques include geometry transformation, colour conversion, adding noise or blurring.

Melexis MLX90640 Far Infrared Thermal Sensor is the sensor that was utilised in the dataset. With a frame rate of roughly 16 frames per second, this inexpensive sensor with 768 FIR pixels produces a low-resolution, privacy-friendly image of  $32 \times 24$  pixels. Take note of how well the image quality captures a fall while yet completely protecting the subject's privacy. It only offers a  $1 \times 8$  pixel image, which is extremely poor resolution. It can measure object temperatures between  $-10^\circ\text{C}$  as well as  $60^\circ\text{C}$ , its operating temperature range is  $0$  to  $50^\circ\text{C}$ . We employ a system of four of these sensors to detect falls and increase the opening range: two at half height and two at ground level. With a baud rate of 115,200, sample rate of 5 Hz, ATMEGA328P microcontroller receives data from four sensors as well as transmits it to an ODROID-C1+ over a UART interface. When temperature detected by the top sensors drops while temperature detected by lower sensors rises, this is known as a fall. This four-sensor system provides data for the dataset at a frame rate of roughly five frames per second.

The model is validated in this work using Berkeley Multimodal Human Action Database (MHAD). The 11 activities are included in MHAD Dataset, an RGB dataset. Seven male and five female individuals complete each action, and the records for each action are identical, resulting in roughly 659 data sequences. Every action in every record of standing, sitting, throwing has five repeats, which adds up to roughly 660 action sequences when action is matched by 82 min of total recording time. Additionally, skeleton and backdrop data are extracted from each subject's T-pose in every record. Some actors execute same number of records in front of, to the right, to the left, and behind the camera. ... others had been photographed in various settings. Chose the video frames at varied angles and distances or speeds to maximise the body position variations because filming human walking is challenging due to its repetitive nature. As a result, a number of scenarios have been used to choose a potential image. In the selection phase, the image frames are visually inspected, and the relevant frames are chosen based on the predetermined criteria.

An in-depth examination of model accuracy and misclassifications is made possible by a confusion matrix, a tool that assesses a classification method performance by showing amount of TP (true positives), TN (true negatives), FP (false positives), FN (false negatives). Suggested model's confusion matrix in Fig. 3. Malicious malware is becoming more common and more sophisticated. In this experiment,

we evaluate a ML classifier that was developed using the Google Colab Python compiler. To assess and measure the detection accuracy of each, the data sets are compared using static analysis to another machine learning classifier that has been trained on other data sets. According to this study, machine learning algorithms are now able to distinguish between safe and dangerous data. Figure 2 usually uses a confusion matrix to calculate these performance indicators. Both expected and actual categories are represented in this matrix. A confusion matrix is commonly used to calculate these performance parameters (see Fig. 2). Both actual and expected classes are covered by this matrix. FP values show how many negative classifications are mistakenly classified as positive, FN values show how many positive classifications are mistakenly classified as negative, TP values show how many positive classifications are correctly classified as positive, TN values show how many negative classifications are correctly classified as negative.

Table 2 shows analysis for thermal infrared and acoustic sensors-based image dataset. Sports multi plant disease detection dataset analysed are SDNET2018 DATASET, Melexis dataset and MHAD DATASET in terms of Correlation coefficient (CC), Detection accuracy, RELIABILITY, Processing time, SCALABILITY.

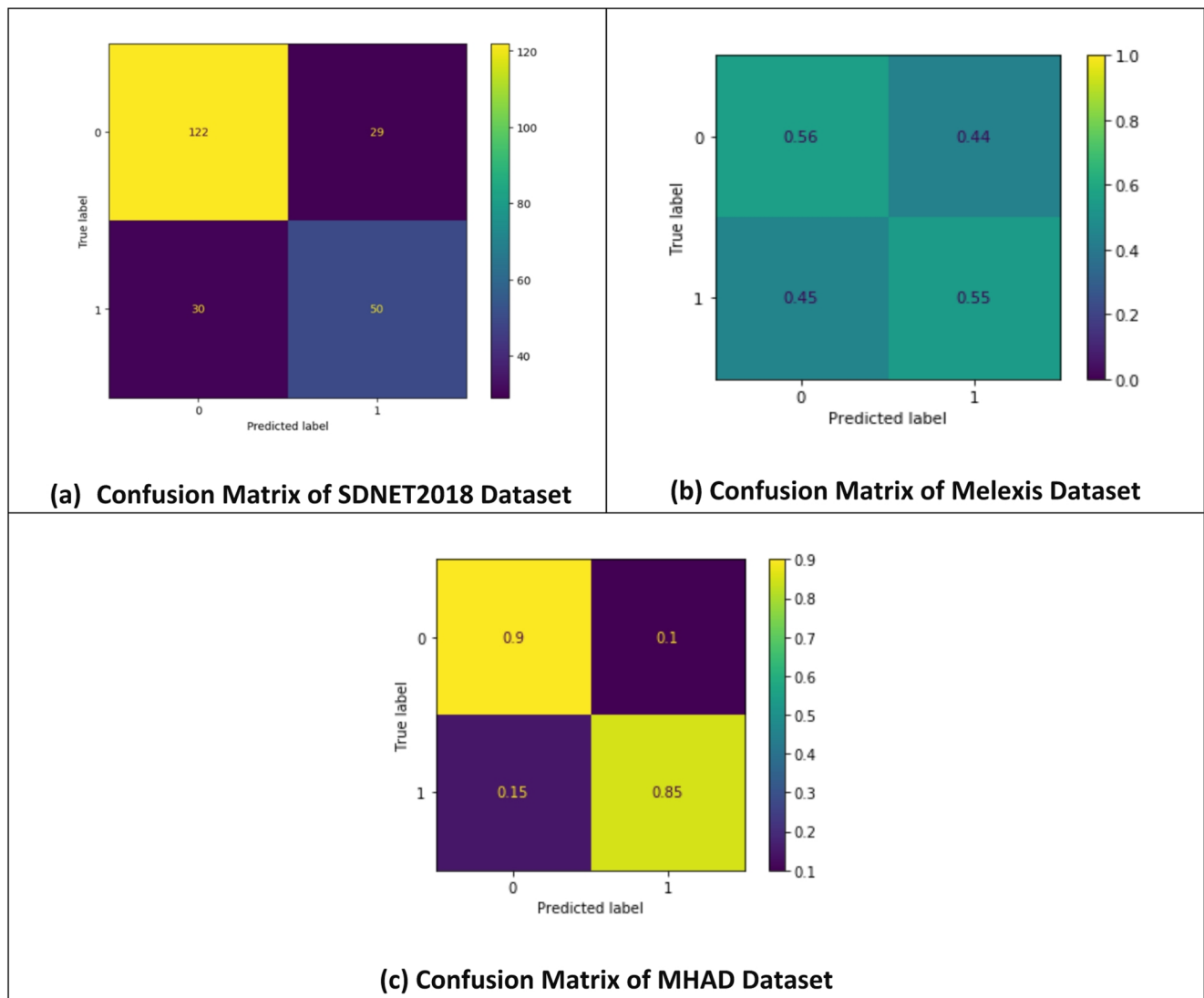
The analysis for Correlation coefficient (CC) is given in Fig. 3. Proposed technique 93% Correlation coefficient (CC), 77% SVM, 89% KNN for SDNET2018 dataset; for Melexis Dataset, proposed technique 96% Correlation coefficient (CC), 75% existing SVM, 79% KNN; for MHAD Dataset, proposed technique 97% Correlation coefficient (CC), 88% SVM, 92% KNN.

Figure 4 gives analysis in Detection accuracy. Proposed technique Detection accuracy 83%, SVM 74%, KNN 80%, for SDNET2018 dataset; for Melexis Dataset proposed technique Detection accuracy 88%, SVM 77%, KNN 82%; proposed technique Detection accuracy 95%, SVM 85%, KNN 90% for MHAD Dataset. Figure 5 shows the comparison of reliability.

RELIABILITY 85%, SVM 73%, KNN 79%; in Melexis dataset, proposed technique RELIABILITY 83%, SVM 70%, KNN 77%; in MHAD dataset, proposed technique RELIABILITY 94%, SVM 69%, KNN 78%.

Processing time analysis is displayed in Fig. 6. Proposed technique 62% processing time, 79% SVM, 81% KNN for SDNET2018 dataset; for Melexis Dataset, proposed technique 60% processing time, 80% SVM, 70% KNN; for MHAD Dataset, proposed technique 58% processing time, 77% SVM, 74% KNN.

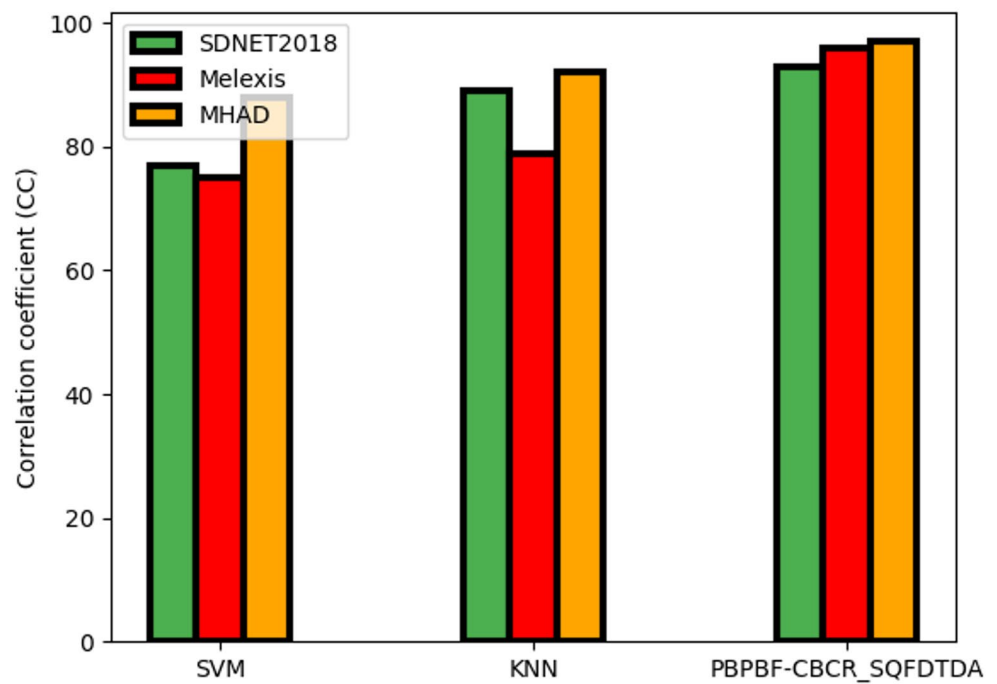
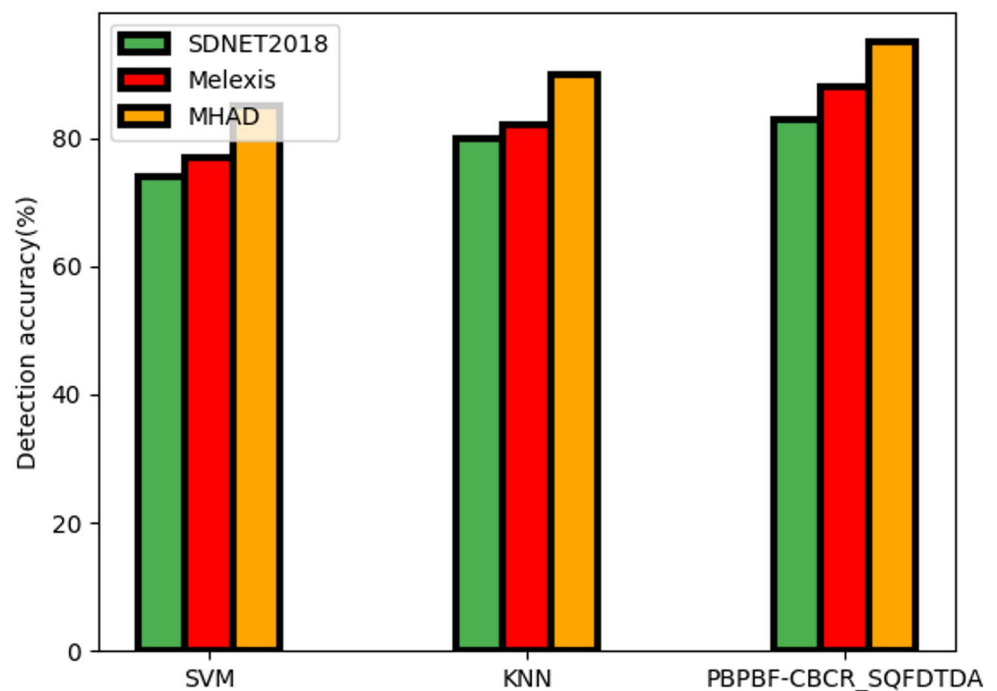
Figure 7 shows analysis in SCALABILITY. Proposed technique SCALABILITY 82%, SVM 72%, KNN 77% for SDNET2018 dataset; for Melexis Dataset proposed technique SCALABILITY 89%, SVM 75%, KNN 82%;



**Fig. 2** Confusion matrix for proposed thermal infrared and acoustic sensors based image dataset **a** SDNET2018 dataset **b** Melexis dataset, **c** MHAD dataset

**Table 2** Comparison between proposed and existing technique for thermal infrared and acoustic sensors-based image dataset

| Dataset           | Techniques         | Correlation coefficient (CC) | Detection accuracy | Reliability | Processing time | Scalability |
|-------------------|--------------------|------------------------------|--------------------|-------------|-----------------|-------------|
| SDNET2018 dataset | SVM                | 77                           | 74                 | 73          | 79              | 72          |
|                   | KNN                | 89                           | 80                 | 79          | 81              | 77          |
|                   | PBPBF-CBCR_SQFDTDA | 93                           | 83                 | 85          | 62              | 82          |
| Melexis dataset   | SVM                | 75                           | 77                 | 70          | 80              | 75          |
|                   | KNN                | 79                           | 82                 | 77          | 70              | 82          |
|                   | PBPBF-CBCR_SQFDTDA | 96                           | 88                 | 83          | 60              | 89          |
| MHAD dataset      | SVM                | 88                           | 85                 | 69          | 77              | 81          |
|                   | KNN                | 92                           | 90                 | 78          | 74              | 85          |
|                   | PBPBF-CBCR_SQFDTDA | 97                           | 95                 | 94          | 58              | 95          |

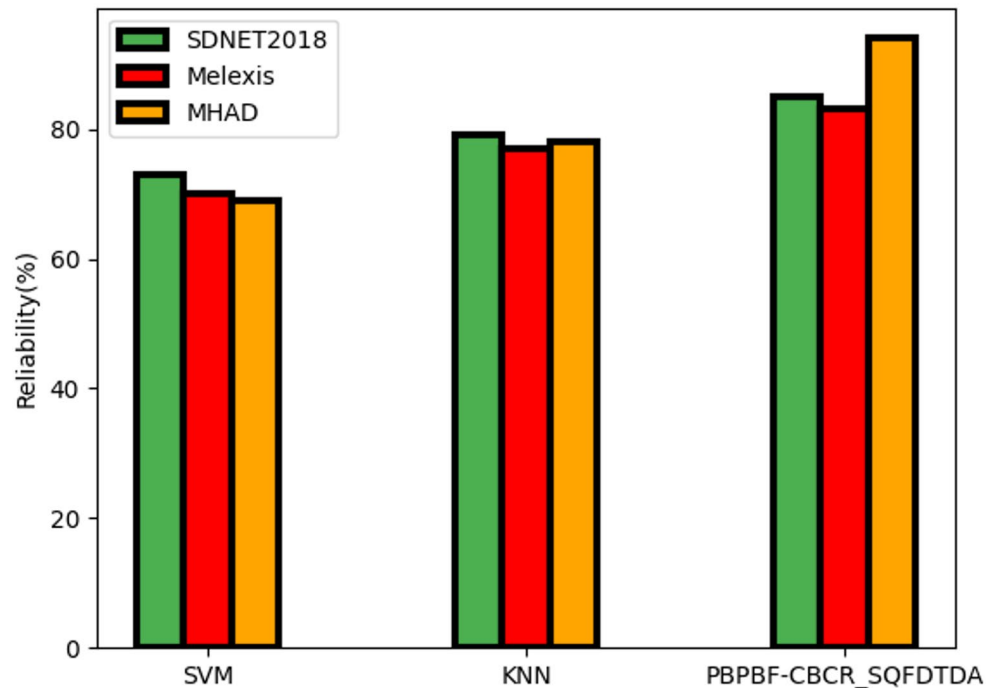
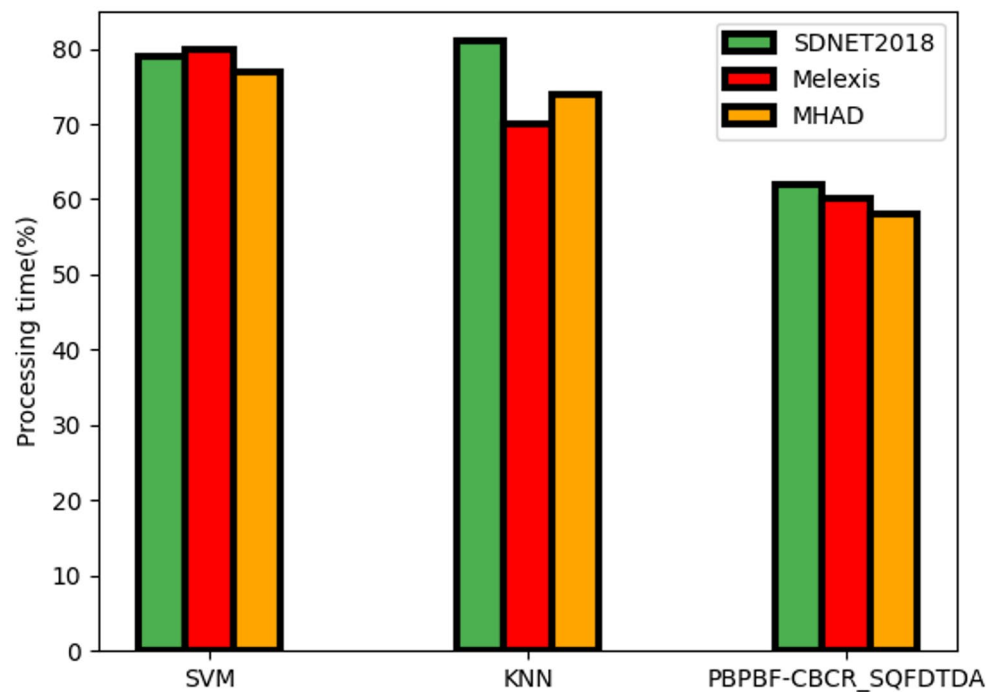
**Fig. 3** Comparison of correlation coefficient (CC)**Fig. 4** Comparison of detection accuracy

proposed technique SCALABILITY 95%, SVM 81%, KNN 85% for MHAD Dataset.

Identifying damage using the Shewhart control chart. In order to identify variations in sensor output from mean or change point, this chart makes use of average values and standard deviation. The mean intervals for healthy and damaged samples, which are based on lower and upper control limits (LCL and UCL), are compared in Fig. 8. Within these bounds, the sensor output is monitored using the Shewhart

chart; any variation from this range may suggest an unusual circumstance. Consequently, a sensor would be deemed irregular if its distribution value changed from  $m$  to  $m + \Delta\sigma$ , as seen in Fig. 8.

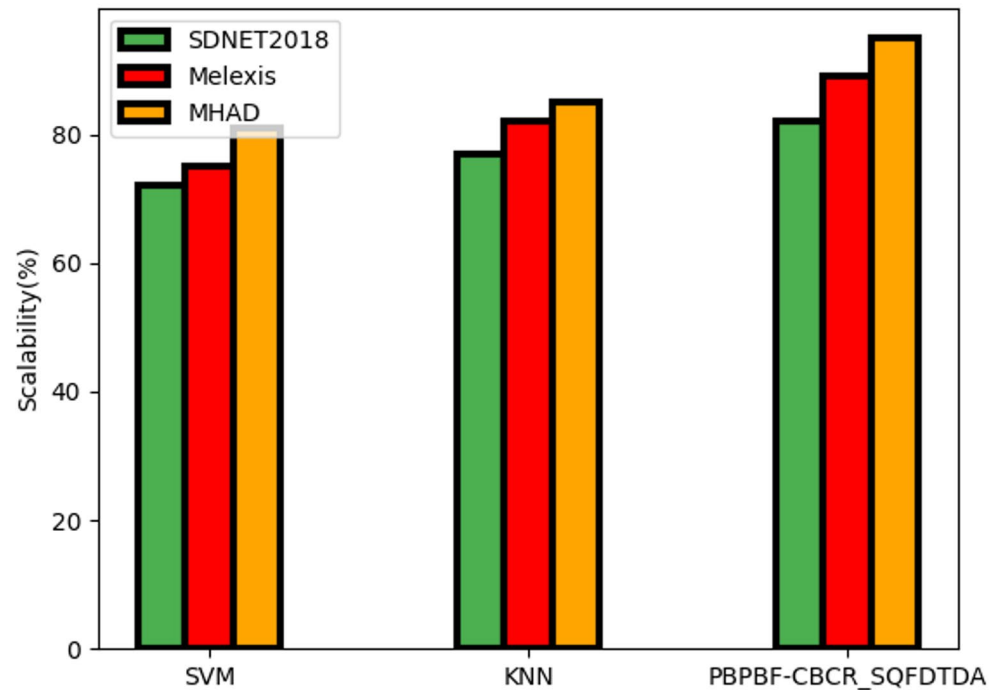
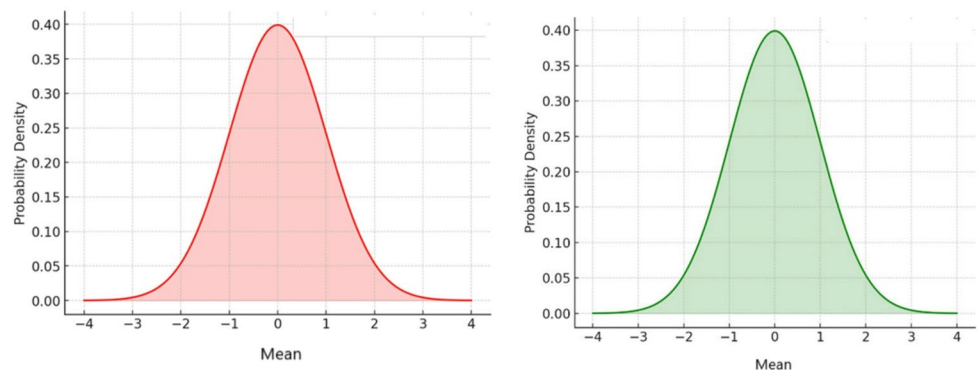
Both LCL and UCL are fully adaptable and follow outputs of the sensors. Different mean and SD values are provided by each sensor, which may lead to different  $\Delta$  and control limit assumptions. Demonstrated, however, that the OoC finding is not affected by the quantity of observations

**Fig. 5** Comparison of reliability**Fig. 6** Comparison of processing time

or a strict control limit. Similar-sized cracks will be arranged logically into bins or ranges. The results of the projected PDF curve demonstrate that, for both surface and length cracks, curve fit corresponded well with crack progression at various levels.

## 6 Discussion

A labelled dataset is utilized to train the chosen ML method, with the labels designating the structural condition (damaged or healthy). A different dataset is then

**Fig. 7** Comparison of scalability**Fig. 8** Probability density belonging to Sensor 3 (square) on left, Sensor 2 (Rectangle) on right, before and after crack development

utilized to validate method to evaluate its generalisation and performance. To get best results, this iterative approach could entail adjusting model's parameters. In the machine learning pipeline for damage detection and SHM, training and validation are essential phases. In these stages, labelled data is utilized to train chosen ML method, and its effectiveness is evaluated to make sure it can effectively generalise to new data. In some applications, thermal and acoustic sensors may be more appropriate than other types of sensors due to their distinct benefits in remote health monitoring. Thermal sensors have the ability to measure body temperature, which is an essential indicator for the diagnosis of many illnesses. A fever, infection, or other medical disorders may be indicated by changes in body temperature. Thermal sensors are perfect for remote health monitoring since they can continuously and non-invasively measure body temperature. Acoustic

sensors can also track respiratory activity, which is another important indicator for disease monitoring and diagnosis. Changes in breathing patterns, for instance, may be a sign of heart failure, asthma, or sleep apnoea. These variations can be recorded by acoustic sensors, which can yield useful information for remote health monitoring. Thermal and acoustic sensors are superior to other types of sensors in terms of sensitivity, accuracy, stretch ability, and flexibility. It is crucial to remember that the particular needs of the health monitoring system will determine which sensor is best. For certain uses or medical circumstances, other sensors could be more appropriate. However, there are certain issues with different kinds of sensors that are utilised in remote patient monitoring systems. For example, problems with data quality, interoperability, health equity, and fairness have been linked to wearable technology, which frequently uses a range of sensors. Furthermore, some



popular fitness trackers have been found to have error margins of up to 25%, raising doubts about the accuracy of the data from these remote patient monitoring technologies. Additionally, some patients cannot afford the equipment that remote patient monitoring systems frequently rely on, such as dependable internet connections. The study acknowledges the use of datasets like SDNET2018, Mel-exis, and MHAD, which were collected under controlled or laboratory conditions; however, it does not explicitly address how the proposed PBPBF-CBCR\_SQFDTDA model would generalize to entirely unseen structures or different environmental contexts, such as varying operational conditions, material types, sensor placements, or climatic factors. The reliance on pre-processed thermal and acoustic data from specific sensor configurations may limit model adaptability, and the absence of cross-dataset validation or stress testing under real-world variabilities suggests potential challenges in scalability and robustness outside the trained scenarios.

## 7 Conclusion

Utilizing ML method, this study suggests a novel technique for improving SHM's scalability for the development of smart healthcare. Here, the probabilistic Bayesian perceptron basis function with convolutional gabor component regression (PBPBF-CBCR) model is used to process the data and detect damage. Next, stacked quadratic finite element decision tree discriminant analysis (SQFDTDA) is used to segment the processed data. The modal peak amplitudes of associated sensors are used as input data to forecast the modal peak amplitude of each sensor. Identifying the system. Integration of ML and SHM performance of method is also analyzed to check reliability into structural health monitoring represents a paradigm shift in way assess as well as manage the integrity of civil infrastructure. By leveraging advanced algorithms, data analysis, and continuous monitoring, machine learning approaches offer a proactive as well as efficient solution for early damage detection and structural health assessment. In conclusion, future of remote patient diagnostics appears bright thanks to the combination of neural networks, inexpensive technology, and non-contact temperature and acoustic sensors. These technologies have the potential to transform healthcare by increasing accessibility and efficiency with additional research and development.

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